

Gen 2.0 Wireless Gigabit Routers – ‘N’ Times Better

Over the past year or so, a new generation of wireless routers—wireless-N and gigabit Ethernet—has taken the market by storm. What makes this generation of wireless routers better than the earlier ones, and what interesting features do these come with?



D-Link DIR-655 Xtreme N gigabit router

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Ever since the Wi-Fi Alliance started testing and certifying products for the 802.11n draft 2.0 specification in June 2007, the market has been greeted by a slew of wireless-N gigabit routers, starting with D-Link's RangeBooster N 650 DIR-655 and Netgear's WNR854T. Now almost every player has a wireless-N gigabit router on offer. What do these mean in the context of wireless routers? The answer to this question is extremely essential in understanding the current

generation of wireless routers, and we shall seek some clarity in this article.

Multimedia-grade Wi-Fi

The 802.11n specification (now in the draft 2.0 stage) is a significant stride ahead for wireless routers, especially those targeted at the SOHO (small office-home office) segment. 802.11n technology enhances the robustness, speed, coverage and other capabilities of Wi-Fi networks. Products based on this standard can deliver up to five times the throughput and twice the range of older wireless products. Therefore the Wi-Fi Alliance rightly

describes an 802.11n network as a 'fat pipe' that multiple users can use effectively across a large area to quickly transfer bulky files and to smoothly run content-rich applications ranging from high-definition video streaming to online gaming. Security options have also been spiked up in the new standard.

The 'n' advantage

What makes 802.11n better than 802.a, b and g? Older technologies tuned out all the weak signals and made use of only the strongest signals detected by the devices. On the other hand, wireless-N products are capable of using echoes to make signals stronger and faster. Hence, they overcome obstructions and reach a wider area easily.

Wireless-N devices are also capable of sending and receiving more than one communication signal at the same time, using multiple antennae. This is what you would find described in product brochures as MIMO (multiple-in, multiple-out) technology. The benefits of MIMO are especially high when all the devices in a network are wireless-N. However, the advantages are definitely noticeable even if only a few devices are MIMO-capable.

Perhaps one of the most noticeable advantages of wireless-N devices, which also makes the technology very multimedia-friendly, is its ability to reduce latency. 802.11n devices have